

a. Call:
lm(formula = totalscore ~ small, data = dfa)

Residuals:

Min	1Q	Median	3Q	Max
-189.44	-50.44	-9.44	43.06	324.38

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	916.442	3.675	249.396	<2e-16 ***
small	12.175	5.369	2.268	0.0236 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared: 0.006608, Adjusted R-squared: 0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363

Interpret the estimates:

$\hat{\beta}_1 = 916.442$: In a regular-sized class, the students' combined aptitude scores (TOTALSCORE), on average, is 916.442

$\hat{\beta}_2 = 12.175$: In a small class, the students' combined aptitude scores (TOTALSCORE), on average, is 12.175 higher than students' in regular-sized class (significant at the 5% significant level).

Conclusion:

According to this regression, the small class seems to improve learning performance in the kindergarten classes.

b. `> summary(m_b_read)`

```
Call:
lm(formula = readscore ~ small, data = dfa)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665 -21.590 -2.665  16.410  194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665    1.488 290.760  < 2e-16 ***
small         6.924     2.174   3.185  0.00151 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295, Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507
```

READSCORE:

$\hat{\beta}_1$: In a regular-sized class, the students' read score, on average, is 432.665.

$\hat{\beta}_2$: In a small class, the student's read score, on average, is 6.924 higher than students' in regular-sized class (significant at the 1% significant level).

Observation:

The improvement in "read" was more evident in the small class.

However, the evidence for improvement in "math" was not strong enough in this sample.

```
> summary(m_b_math)

Call:
lm(formula = mathscore ~ small, data = dfa)

Residuals:
    Min       1Q   Median       3Q      Max
-144.777 -35.028  -5.028   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777    2.449 197.545  <2e-16 ***
small         5.251     3.578   1.467   0.143
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427
```

MATHSCORE:

$\hat{\beta}_1$: In a regular-sized class, the students' math score, on average, is 483.665.

$\hat{\beta}_2$: In a small class, the student's math score, on average, is 5.251 higher than students' in regular-sized class. However, it is **not** statistically significant at the 10% significant level.

c. Call:
lm(formula = totalscore ~ aide, data = dfc)

Residuals:

	Min	1Q	Median	3Q	Max
	-191.748	-50.442	-5.442	40.558	312.558

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	916.442	3.559	257.529	<2e-16 ***
aide	4.306	4.994	0.862	0.389

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom

Multiple R-squared: 0.0008898, Adjusted R-squared: -0.0003068

F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888

Interpret the estimates:

$\hat{\beta}_1$: In a regular-sized class, the students' combined aptitude scores (TOTALSCORE), on average, is 916.442.

$\hat{\beta}_2$: In a regular-sized class with a teacher aide, the students' combined aptitude scores (TOTALSCORE), on average, is 4.306 higher than students' in regular-sized class. However, it is **not** statistically significant at the 10% significant level.

Conclusion:

There is no evidence that adding teaching assistants significantly improves learning outcomes in this sample.

d.

> summary(m_d_read)

Call:

lm(formula = readscore ~ aide, data = dfc)

Residuals:

Min	1Q	Median	3Q	Max
-74.665	-21.536	-2.536	15.464	194.335

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	432.665	1.460	296.341	<2e-16 ***
aide	2.871	2.049	1.401	0.161

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom

Multiple R-squared: 0.002347, Adjusted R-squared: 0.001152

F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615

READSCORE:

$\hat{\beta}_1$: In a regular-sized class, the students' read score, on average, is 432.665.

$\hat{\beta}_2$: In a regular-sized class with a teacher aide, the students' read score, on average, is 2.871 higher than students' in regular-sized. However, it is **not** statistically significant at the 10% significant level.

Conclusion :

There is no evidence that adding a teacher aide significantly improves reading and math performance in this sample.

> summary(m_d_math)

Call:

lm(formula = mathscore ~ aide, data = dfc)

Residuals:

Min	1Q	Median	3Q	Max
-146.212	-31.212	-5.777	29.223	142.223

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.777	2.355	205.464	<2e-16 ***
aide	1.435	3.304	0.434	0.664

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.79 on 835 degrees of freedom

Multiple R-squared: 0.0002258, Adjusted R-squared: -0.0009715

F-statistic: 0.1886 on 1 and 835 DF, p-value: 0.6642

MATHSCORE:

$\hat{\beta}_1$: In a regular-sized class, the students' math score, on average, is 483.777.

$\hat{\beta}_2$: In a regular-sized class with a teacher aide, the students' math score, on average, is 1.435 higher than students' in regular-sized. However, it is **not** statistically significant at the 10% significant level.

Code:

```
load("/Users/liaochenghao/Desktop/Financial Econometrics/Homework/HW1/Data/star5_small.rdata")
df <- star5_small
names(df)

# (a)
dfa <- subset(df, aide == 0)
m_a <- lm(totalscore ~ small, data = dfa)
summary(m_a)

# (b)
m_b_read <- lm(readscore ~ small, data = dfa)
m_b_math <- lm(mathscore ~ small, data = dfa)

summary(m_b_read)
summary(m_b_math)

# (c)
dfc <- subset(df, small == 0)
m_c <- lm(totalscore ~ aide, data = dfc)
summary(m_c)

# (d)
m_d_read <- lm(readscore ~ aide, data = dfc)
m_d_math <- lm(mathscore ~ aide, data = dfc)

summary(m_d_read)
summary(m_d_math)
```